

J.K.K.N COLLEGE OF ARTS & SCIENCE

(AUTONOMOUS)

(ACCREDITED BY NAAC, APPROVED BY AICTE,
AFFILIATED TO PERIYAR UNIVERSITY, SALEM-11)

KOMARAPALAYAM- 638 183, NAMAKKAL (Dt).



B.Sc., MATHEMATICS PROGRAMME

[Choice Based Credit System (CBCS)]

OBE REGULATIONS AND SYLLABUS FRAMEWORK

(With effect from the academic year 2025-2026 onwards)

Introduction

B.Sc. Mathematics

Mathematics is the study of quantity, structure, space and change, focusing on problem solving, with wider scope of application in science, engineering, technology, social sciences etc. The key core areas of study in Mathematics include Algebra, Analysis (Real & Complex), Differential Equations, Geometry, and Mechanics. The Bachelor's Degree B.Sc. Mathematics is awarded to the students on the basis of knowledge, understanding, skills, attitudes, values and academic achievements expected to be acquired by learners at the end of the Programme. Learning outcomes of Mathematics are aimed at facilitating the learners to acquire these attributes, keeping in view of their preferences and aspirations for gaining knowledge of Mathematics.

Bachelor's degree in Mathematics is the culmination of in-depth knowledge of algebra, calculus, geometry, differential equations and several other branches of Mathematics. This also leads to study of related areas like Computer science, Financial Mathematics, Statistics and many more. Thus, this programme helps learners in building a solid foundation for higher studies in Mathematics. The skills and knowledge gained have intrinsic aesthetics leading to proficiency in analytical reasoning. This can be utilised in Mathematical modelling and solving real life problems.

Students completing this programme will be able to present Mathematics clearly and precisely, make abstract ideas precise by formulating them in the language of Mathematics, describe Mathematical ideas from multiple perspectives and explain fundamental concepts of Mathematics to non-Mathematicians.

Completion of this programme will also enable the learners to join teaching profession, enhance their employability for government jobs, jobs in banking, insurance and investment sectors, data analyst jobs and jobs in various other public and private enterprises.

Definitions:

Programme : A course of study leading to the award of a degree in a discipline.

Course : Course refers to the subject offered under the degree programme.

Part I : Tamil/other languages

Part II : English language

Part III : (1) The core courses related to the programme concerned and practical.

(2) Elective generic/discipline specific courses related to the core courses of the programme concerned.

Part IV : Non-Major Elective means elective subjects offered by other Major Departments and are to be selected by the students. Skill Enhancement Courses means the courses offered as skill-based courses aimed at imparting Advanced Skill. Value Education (offered in the III year - V Semester) and Environmental Studies (offered in the II year - III & IV Semesters), Internship / Industrial Training (end of IV Semester – 2 Weeks) are offered.

Part V

Extension Activities:

All those activities under NSS/ NCC/ Sports/ YRC programme and other co and extracurricular activities.

A detailed explanation of the above with relevant credits are given under Scheme of Examination along with Distribution of Marks and Credits.

Duration : It refers to the stipulated years of study to complete a programme as prescribed by the University from time to time. Currently for the undergraduate programme the duration of study is THREE years. These regulations shall apply to the regular course of study in approved institutions of the University.

Credits : It refers to the weightage given to each course of study (subject) by the experts of the Board of Studies of Mathematics.

Credit System : It refers to the course of study under this regulation, where weightage of credits is spread over to different semesters during the period of study and the Cumulative Grade Point Average shall be awarded based on the credits earned by the students. A total of 140 credits with 4 extra credits (2 credits for certificate courses and 2 credits for Swayam, MOOC, NPTEL etc.,) are prescribed for the Undergraduate Programme (Three years).

Choice-Based Credit System: All Undergraduate Programmes offered by the College shall be under Choice Based Credit System (CBCS). This is to enhance the quality and mobility of the students within and between the Universities in the country and abroad.

Eligibility for Admission to the Course:

Candidates seeking admission to first year of the Bachelor of Science - Mathematics is required to have passed the higher secondary examination with Mathematics conducted by the Government of Tamil Nadu or an examination accepted as equivalent there to.

Duration of the Course:

The course shall extend over a period of three years comprising six semesters with two semesters in one academic year. There shall not be less than 90 working days for each semester. Examination shall be conducted at the end of every semester for the respective subjects.

Course of Study:

The course of study for the UG degree courses of all branches consists the following:

Part - I: Language

Tamil or any one of the following modern/classical languages i.e. Telugu, Kannada, Malayalam, Hindi, Sanskrit, French, German, Arabic & Urdu.

The subject shall be offered during I, II, III, IV semesters with one examination at the end of each semester (12 credits).

Part – II: English

The subject shall be offered during I, II, III, IV semesters with one examination at the end of each semester (12 credits).

Part – III:**Core subjects:**

As prescribed in the scheme of examination.

Elective courses:

Elective subjects are to be selected from the list of electives prescribed by the Board of Studies concerned.

Part – IV Comprises of 1) Skill Enhancement courses, 2) Environmental Studies, 3) Value Education, 4) Internship / Industrial Training.

Part – V Extension Activities

Every student shall participate compulsorily for the period of not less than two years (4

semesters) in any one of the following programmes NSS, NCC, Sports., YRC.

Examination:

To be conducted at the end of each semester. For Environmental Studies and Practical refer Scheme of Examination.

Eligibility to appear for the examinations:

A Candidate shall be permitted to appear for the end semester examinations for any semester (practical/theory) if he/she secures not less than 75% of attendance in the number of working days during the semester.

A candidate who secures attendance from 65% to 74% has to pay a fine of Rs.800/- to appear for the examination.

A candidate who secures attendance from 50 % to 64% has to pay a fine of Rs.800/- and can appear for both semester papers together attendance the end of the later semester.

A candidate who has secured less than 50% of attendance in any semester shall not be permitted to appear for the regular examinations and has to continue the study in the subsequent semester. He/she has to rejoin the semester in which the attendance is less than 50%.

Restrictions to appear for the examinations:

Any candidate having arrear paper(s) shall have the option to appear in any arrear paper along with the regular semester papers.

Candidates who fail in any of the course of Part I, II, III, IV & V of UG degree examinations shall complete the course concerned within 5 years from the date of admission to the said programme, and should they fail to do so, they shall take up the examination in the texts/ revised syllabus prescribed for the immediate next batch of candidates. If there is no change in the texts/syllabus they shall appear for the examination in that course with the syllabus in vogue until there is a change in the texts or syllabus. In the event of removal of that course consequent to change of regulation and / or curriculum after 5-year period, the candidates shall have to take up an equivalent course in the revised syllabus as suggested by the Chairman and fulfill the requirements as per the regulations of the curriculum for the award of the degree.

Medium of Instruction and examinations:

The medium of instruction and examinations for the courses of Part I & II shall be the language concerned. For part III & IV, the medium of instruction is English.

Submission of Record Note Books for practical examinations:

Candidates appearing for practical examinations should submit bonafide record note books prescribed for practical examinations, otherwise the candidates shall not be permitted to appear for the practical examinations.

Passing Minimum & Mark Distribution:

The following tables are the distribution of marks for external and internal for End Semester (external) Examination and continuous internal assessment and passing minimum marks for theory papers of UG programmes.

End Semester Examination

Total Marks	External		Internal		Overall Passing Minimum for total marks (Internal + External)
	Max. Marks	Passing Minimum for external alone	Max. Marks	Passing minimum for internal alone	
100	60	24	40	16	40

Table – 1(B): The following are the distribution of marks for continuous internal assessment in theory papers of UG programmes:

Summative Assessment:

S.No.	For Theory - UG Courses	Distribution of Marks
1	Tests	10
2	Assignments (2 Nos.)	
3	Attendance	5
	Total Marks	40

Formative

TEST PORTION ALLOCATION FOR CIA & MODEL EXAMINATIONS:

Test No.	Portion	Total duration	Max. marks
CIA-I	2 units	1 hour	30
CIA-II	2 units	1 hour	30
Model examination	5 units	3 hours	75

WEIGHTAGE OF INTERNAL MARKS:

Test No.	Max marks	Weightage	Total
CIA-I	30	15	25
CIA-II	30		

Model examination	75		
Activity*	5	5	
Attendance	5	5	

Activity: Assignment/Case study/Field survey/PPT presentation/group discussion/Subject Viva-voce/Report Writing/Content Writing/Survey studies/Mind map/Flow chart preparation/Model making/Review of report/Debate/Group Discussion/Surprise test/Open book test etc.

QUESTION PAPER PATTERN FOR CONTINUOUS INTERNAL ASSESSMENT (30 Marks)

Section	Bloom's category	Marks	Total
A (5X1) Multiple Choice Questions	K1, K2 (Remember & Understand)	5	30
B (1X5) Open choice question (One out of Two)	K3, K4 (Apply and Analyze)	5	
C (2X10) Either or Questions	K5, K6 (Evaluate and Creative)	20	

QUESTION PAPER PATTERN FOR END SEMESTER EXAMINATIONS

A. WRITTEN ASSESSMENT – 40 MARKS

Section	Question Type	Fink's Dimension	Marks Description
A	Objective & Short Answers (5 × 2 marks)	Foundational Knowledge	10
B	Numerical Problem Solving (2 × 5 marks)	Application	10
C	Essay/Long Answer: Theory + Diagram/Derivation	Integration	10
D	Reflection: Physics in Daily Life / Ethical Science	Human Dimension + Caring	10
Total			40 Marks

B. ORAL ASSESSMENT / VIVA – 20 MARKS

Component	Fink's Dimension	Marks Description
Viva on Experiment / Concept	Application + Foundational	10
Conceptual Q&A / Cross-linking	Integration	5
Communication and	Learning How to Learn	5

Reflection		
Total		20 Marks

Table – 1(C): The following are the distribution of marks for the continuous internal assessment in UG practical courses:

S.No.	For Practical - UG Courses	Distribution of Marks
1	Tests	25
2	Observation Notebook	10
3	Attendance	5
	Total Marks	40

MARK ALLOTMENT FOR MODEL & END SEMESTER PRACTICAL EXAMINATIONS:

Internal	External	Total Marks	Passing minimum in Internal	Passing minimum in ESE
40	60	100	16	24

Question paper pattern will be decided by the Board of Studies of Mathematics

BREAKUP MARKS FOR ATTENDANCE:

90 % and above	-	5 marks
85 % - 89 %	-	4 marks
75 % - 84 %	-	3 Marks
Less than 75%	-	2 Marks

(As a special case under Medical / Sports / other activities, attendance exemption maybe given by the Principal and the attendance marks may be awarded). Continuous Internal Assessment (CIA) as furnished in the syllabus. The marks shall be submitted to the COE by the respective departments. A Candidate who passes the examination in all the courses of Part I, II, III, IV & V shall be declared to have passed the whole examination. The syllabus for various courses shall be clearly demarcated into five viable units in each paper/subject.

Revision of Regulations and Curriculum:

The above Regulation and Scheme of Examinations shall be in vogue without any change for a minimum period of three years from the date of approval. The College may revise / amend the Regulations and Scheme of Examinations, if found necessary.

XV) Internship (2 weeks):

The students have to undergo a training for a period of two weeks at the end of the semester IV during vacation. The Student can select any organization such as Government / private like industry, R&D organizations, scientific companies, etc., in consultation with the staff coordinator and Head of the Department. They must maintain a work diary and prepare a report of the training undergone and submit the same. At the end of the fifth semester, viva-voce will be conducted on the basis of the report submitted by the student. The viva-voce shall be conducted by the internal examiner.

The student is awarded as commended / Highly Commended.

Credit distribution:

Credit distribution for B.Sc Mathematics under Learning Outcome based Curriculum framework – Choice Based Credit System Pattern with effect from 2024-2025 Onwards.

Credit Distribution Based on TANSCH/UGC Guidelines

Part	Subjects	No. of Credits
PART-I	Language	12
PART-II	English	12
PART-III	Core Courses & Elective Courses[in Total]	92
PART-IV	Skill Enhancement Courses	17
	Environmental Studies	2
	Value Education	2
	Internship / Industrial Training	2
PART-V	Extension Activity	1
Grand Total		
Extra Credit (if any)		
	Total No. of Subjects	42
	Total credits	140

Consolidated Semester wise and Component wise Credit distribution

Parts	Sem I	Sem II	Sem III	Sem IV	Sem V	Sem VI	Total Credits
Part I	3	3	3	3	-	-	12
Part II	3	3	3	3	-	-	12
Part III	13	13	13	13	22	18	92
Part IV	4	4	3	6	4	2	23
Part V	-	-	-	-	-	1	1
Total	23	23	22	25	26	21	140

*Part I, II, and Part III components will be separately taken into account for CGPA (Cumulative Grade Point Average) calculation and classification for the under graduate programme and the other components. IV, V have to be completed during the duration of the programme as per the norms, to be eligible for obtaining the UG degree.

Credit Distribution for UG Programme Semester wise

Sem I	Credit	H	Sem II	Credit	H	Sem III	Credit	H	Sem IV	Credit	H	Sem V	Credit	H	Sem VI	Credit	H
Part 1. Language – Tamil	3	6	Part..1. Language – Tamil	3	6	Part..1. Language – Tamil	3	6	Part..1. Language – Tamil	3	6	5.1 Core Course –\CC IX	4	5	6.1 Core Course – CC XIII	5	6
Part.2 English	3	6	Part..2 English	3	6	Part..2 English	3	6	Part..2 English	3	6	5.2 Core Course – CC X	4	5	6.2 Core Course – CC XIV	4	6
1.3 Core Course – CC I	4	4	2..3 Core Course – CC III	4	4	3.3 Core Course – CC V	4	4	4.3 Core Course – CC VII Core Industry Module	3	3	5. 3.Core Course CC -XI	4	4	6.3 Core Course – CC XV	4	6
1.4 Core Course – CC II	4	4	2.4 Core Course – CC IV	4	4	3.4 Core Course – CC VI	4	4	4.4 Core Course – CC VIII	4	4	5. 4.Core Course –/ Project with viva- voce CC -XII	4	4	6.4 Elective -VII Generic/ Discipline Specific	3	5
1.5 Elective I Generic/ Discipline Specific	5	6	2.5 Elective II Generic/ Discipline Specific	5	6	3.5 Elective III Generic/ Discipline Specific	5	6	4.5 Elective IV Generic/ Discipline Specific	6	6	5.5 Elective V Generic/ Discipline Specific	3	5	6.5 Elective VIII Generic/ Discipline Specific	3	5
1.6 Skill Enhancement Course NMC-1	2	2	2.6 Skill Enhancement Course NMC-2	2	2	3.6 Skill Enhancement Course SEC-3, (Entrepreneurial Skill)	1	1	4.6 Skill Enhancement Course SEC-5	2	2	5.6 Elective VI Generic/ Discipline Specific	6	5	6.6 Extension Activity	1	-
1.7 Skill Enhancement -(Foundation Course) – SEC -1	2	2	2.7 Skill Enhancement Course –SEC-2	2	2	3.7 Skill Enhancement Course SEC-4	2	2	4.7 Skill Enhancement Course SEC-6	2	2	5.7 Value Education	2	2	6.7 Professional Competency Skill	2	2
						3.8 E.V.S.	-	1	4.8 E.V.S	2	1	5.8 Summer Internship /Industrial Training	2	-			
	23	30		23	30		22	30		25	30		26	30		21	30
Total – 140 Credits																	

B. Sc., Mathematics

Programme Outcomes:

PO1: Deep Disciplinary Knowledge & Critical Thinking: Gain a thorough understanding of the chosen field, coupled with strong critical thinking skills; and this empowers the learners to analyze information, evaluate arguments, identify biases, and solve non-routine problems effectively.

PO2: Analytical Reasoning & Scientific Inquiry: Master the ability to critically evaluate data from diverse sources, draw sound conclusions based on evidence, and approach problem-solving with a scientific mindset.

PO3: Independent Learning & Lifelong Development: Cultivate the skills to work independently, manage projects effectively and continuously acquire new knowledge and skills through self-directed learning; and this program equips the learners for lifelong learning and adaptation in a dynamic world.

PO4: Communication & Collaboration: Develop strong communication and collaboration skills, allowing the learners to effectively communicate complex ideas, work cohesively in teams, and learn from diverse perspectives.

PO5: Ethical Reasoning & Professionalism: Foster a strong sense of ethical responsibility, conducting research and professional activities with integrity and fairness; and this program prepares the learners to be a valuable and ethical contributors to their chosen field.

Programme Specific Outcomes

PSO1: Acquire good knowledge and understanding, to solve specific theoretical & applied problems in different area of mathematics & statistics.

PSO2: Understand, formulate, develop mathematical arguments, logically and use quantitative models to address issues arising in social sciences, business and other context /fields.

PSO3: To prepare the students who will demonstrate respectful engagement with other's ideas, behaviors, beliefs and apply diverse frames of references to decisions and actions.

PSO4: To create effective entrepreneurs by enhancing their critical thinking, problem solving, decision making and leadership skill that will facilitate startups and high potential organizations.

PSO5: Learn independently and improve one's performance. Students are equipped to appear competitive examinations.

Mapping of Course Learning Outcomes (CLOs) with Programme Outcomes (POs) and Programme Specific Outcomes (PSOs) can be carried out accordingly, assigning the appropriate level in the grids:(put tick mark in each row)

PO/PSO	PSO1	PSO2	PSO3	PSO4	PSO5
PO1	✓				
PO2		✓			
PO3			✓		
PO4				✓	
PO5					✓

B.Sc., Mathematics Programme:
SCHEME OF LEARNING AND EVALUATION
 (For the students admitted in the academic year 2025 – 2026 onwards)
 Outcome Based Curriculum Framework with CBCS

FIRST SEMESTER										
Part	Course Code	Title of the Course	Exam (Hrs)	Credits	Hours			Max. Marks		
					L	P	Total	INT	EXT	Total
I	25UGTA01	General Tamil - I	3	3	5	-	5	40	60	100
II	25UGEN01	General English - I	3	3	5	-	5	40	60	100
III	25UMAC01	Algebra and Trigonometry	3	4	5	-	5	40	60	100
III	25UMAC02	Differential Calculus	3	4	5	-	5	40	60	100
III		Paper – I – Allied Physics-I	3	4	4	-	4	40	60	100
		Allied Physics-Practical - I/ NSS-I/Sports-I/First Aid & Safety/Swayam	3	1	-	2	2	40	60	100
IV	25UMANM1	NMC – 1 – Quantitative Aptitude - I	3	2	2	-	2	40	60	100
IV	25UMAS01	SEC – I – Financial Mathematics	3	2	2	-	2	40	60	100
	TOTAL			23	28	2	30			
SECOND SEMESTER										
I	25UGTA02	General Tamil - I	3	3	5	-	5	40	60	100
II	25UGEN02	General English - II	3	3	5	-	5	40	60	100
III	25UMAC03	Analytical Geometry (Two & Three Dimensions)	3	4	5	-	5	40	60	100
III	25UMAC04	Integral Calculus	3	4	5	-	5	40	60	100
III		Paper – II – Allied Physics-II	3	4	4	-	4	40	60	100
		Allied Physics-Practical - II/ NSS-II/Sports-II/First Women;s Rights/Swayam	3	1	-	2	2	40	60	100
IV	25UMANM2	NMC – 2 - Quantitative Aptitude - II	3	2	2	-	2	40	60	100
IV	25UMAS02	SEC – 2 – Office Automation Practical	3	2	2	-	2	40	60	100
	TOTAL			23	28	2	30	-	-	

THIRD SEMESTER

Part	Course Code	Title of the Course	Exam (Hrs)	Credits	Hours			Max. Marks		
					L	P	Total	INT	EXT	Total
I	25UTAF03	Tamil – III	3	3	5	-	6	40	60	100
II	25UENF03	English - III	3	3	5	-	6	40	60	100
III	25UMAC05	Vector Calculus and its Applications	3	3	3	-	3	40	60	100
III	25UMAC06	Differential Equations and its Applications	3	4	4	-	4	40	60	100
III		Paper – I - Theory	3	4	4	-	4	40	60	100
		Practical	3	1	-	2	2	40	60	100
		NSS-III/Sports-III/Human Rights/Swayam								
IV	25UMAS03	SEC – 3 Statistics with Excel Programming	3	2	2	-	2	40	60	100
IV	25UMAS04	NME – III Quantitative Aptitude - III	3	2	2	-	2	40	60	100
IV	25UEVS01	Environmental Studies	3	-	1	-	1	-	-	-
	TOTAL			22	28	2	30	-	-	700
FOURTH SEMESTER										
I	25UTAF04	Tamil – IV	3	3	5	-	6	40	60	100
II	25UENF04	English - IV	3	3	5	-	6	40	60	100
III	25UMAC07	Industrial Statistics	3	3	4	-	4	40	60	100
III	25UMAC08	Elements of Mathematical Analysis	3	4	4	-	4	40	60	100
III	Elective Course – II	Paper – II - Theory	3	4	4	-	4	40	60	100
		Practical	3	1	-	2	2	40	60	100
		NSS-IV/Sports-IV/Human Psychology/Swayam								
IV	25UMAS05	NME –IV - Quantitative Aptitude - IV	3	2	-	3	3	40	60	100
	25UMASP01	SEC – IV - Latex Practical	3	2				40	60	
IV	25UEVS01	Environmental Studies	3	2	1	-	1	40	60	100
	Total			24	25	5	30			800

Elective / Allied Mathematics

Name of the Course	Paper Code
Paper I - Allied Mathematics - I	25UMAGE1
Paper II - Allied Mathematics - II	25UMAGE2
Allied Mathematics - Practical*	25UMAGEP01

*Examination at the end of the II - Semester.

**No Examination - Participation in NCC/ NSS/ RRC/ YRC/ Others if any.

Elective Courses Generic Specific for All Computer Science Departments

Name of the Course	Paper Code
Theory[#]	
Discrete Mathematics - I	25UMAGE3
Discrete Mathematics - II	25UMAGE4
Numerical Methods	25UMAGE5
Optimization Techniques	25UMAGE6
Introduction to Linear Algebra	25UMAGE7
Graph Theory and its Applications	25UMAGE8
Numerical Methods - I	25UMAGE9
Numerical Methods - II	25UMAGE10
Practicals[*]	
Discrete Mathematics	25UMAGEP02
Numerical Methods	25UMAGEP03
Optimization Techniques	25UMAGEP04
Introduction to Linear Algebra	25UMAGEP05
Graph Theory and its Applications	25UMAGEP06

GUIDELINES FOR ASSESSMENT:

The course for the degree of Bachelor of Science in Computer Science (Artificial Intelligence and Data Science) consists of three academic years and divided into six semesters.

Courses and their credit values will be decided by the respective board.

These regulations shall take effect from the academic year 2025-26, i.e., for students who are to be admitted to the first year of the course during the academic year 2025-26 and thereafter.

The theory examinations shall be three hours duration to each paper at the end of each semester.

Practical examinations for undergraduate (UG) courses will be 3/4/6 hours and conducted in the even semesters, namely the second, fourth and sixth semesters or according to their BoS specifications.

In case students have arrears in practical examinations they must apply during the respective semesters.

The Candidate failing in any subject(s) will be permitted to appear for each failed subject(s) in the subsequent examinations.

Elective course one from group A and one from group B will be offered in the corresponding semesters.

Marks for theory/practical/project courses should be out of 100.

The learners must complete their 15 days internship at the end of the fourth semester vocation leave and examination will be conducted on fifth semester.

Summer Internship course will be assessed by Commended or Highly commended.

The extension activity will be completed in the sixth semester. Assessment for this activity will be based on credits instead of marks, with only one credit provided.

In addition, seminars, assignments and personality development programmes such as NCC, NSS, YRC, Sports and Games and extension activities shall be offered.

If a candidate fails to submit the report on project work on or before the date specified by the College, he/she shall be deemed to have failed in the project work and shall submit the report for evaluation in the next even semester only.

Extra credits will be added for Swayam/MOOC courses. Massive Open Online Course (MOOC) should be completed at any time of the Programme. The respective Board shall identify and prescribe a list of relevant MOOCs, available on SWAYAM / NPTEL / any other MOOC portal. The course must be of minimum 4 weeks duration.

The final assessments will be based on CIA (Continuous Internal Assessment) I, II and ESE (End Semester Examinations).

Core Paper – I	ALGEBRA & TRIGONOMETRY		
Course code	Total Hours	Contact Hours	Credits
25UMAC 01	4	4	4
	Course Objectives		
	The main objectives of this course are		
1	Basic ideas on the Theory of Equations, Matrices and Number Theory.		
2	Knowledge to find expansions of trigonometry functions, solve Theoretical and applied problems.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Classify and Solve reciprocal equations	K1 K2 K3 K4 K5 K6	
CLO 2	Find the sum of binomial, exponential and logarithmic series		
CLO 3	Find Eigen values, eigen vectors, verify Cayley - Hamilton theorem and diagonalize a given matrix		
CLO 4	Expand the powers and multiples of trigonometric functions in terms of sine and cosine		
CLO 5	Determine relationship between circular and hyperbolic functions and the summation of trigonometric series		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Reciprocal Equations - Standard form - Increasing or decreasing the roots of a given equation-Removal of terms, Approximate solutions of roots of polynomials by Horner’s method–related problems. (Book 1 - Chapter 6:Sections16,17,19,30).		
Unit-II	Summation of Series:Binomial - Exponential – Logarithmic Series (Theorems without proof) - Approximations - related problems. (Book 1 - Chapter 3: Sections 10,14; Chapter 4: Sections -1,2,3,5,7,8,9,11).		
Unit-II I	Inverse of a square matrix upto order 3, Characteristic equation - Eigen values and Eigen Vectors - Similar matrices - Cayley - HamiltonTheorem(Statementonly) - Finding powers of square matrix, Diagonalization of square matrices - related problems. (Book 2 - Chapter 2: Sections - 8,16).		
Unit-IV	Expansions of $\sin^n\theta$, $\cos^n\theta$ in powers of $\sin\theta$, $\cos\theta$ - Expansion of $\tan^n\theta$ interms of $\tan\theta$, Expansions of $\cos^n\theta$, $\sin^n\theta$, $\cos^m\theta\sin^n\theta$ - Expansions of $\tan(\theta_1+\theta_2+,...,\theta_n)$ - Expansions of $\sin\theta,\cos\theta$ and $\tan\theta$ in terms of θ - related problems. (Book 3 - Chapter 3: Sections1 to 5).		

Unit-V	Hyperbolic functions - Relation between circular and hyperbolic functions Inverse hyperbolic functions, Logarithm of complex quantities, Summation of trigonometric series - related problems. (Book3 - Chapter 4; Chapter 5; Chapter 6: Sections 1,3,3.1 Related problems.)
Text Books	1. Manickavasagam Pillai, T.K., T. Natarajan and Ganapathy KS - Algebra Vol-I, Viswanathan Publishers and Printers Pvt. Ltd., -2008. 2. Manickavasagam Pillai, T.K., T. Natarajan and Ganapathy KS - Algebra Vol-II, Viswanathan Publishers and Printers Pvt Ltd., -2008. 3. Manichavasagam Pillai, T.K. and S. Narayanan, Trigonometry - Viswanathan Publishers and Printers Pvt. Ltd. 2013.
Reference Books	1. W.S. Burnstine and A.W. Panton, Theory of equations 2. David C. Lay, Linear Algebra and its Applications, 3 rd Ed., Pearson Education Asia, Indian Reprint, 2007. 3. G.B. Thomas and R.L. Finney, Calculus, 9 th Ed., Pearson Education, Delhi, 2005. 4. C.V. Durell and A. Robson, Advanced Trigonometry, Courier Corporation, 2003. 5. J. Stewart, L. Redlin, and S. Watson, Algebra and Trigonometry, Cengage Learning, 2012. 6. Calculus and Analytical Geometry, G.B. Thomas and R.L. Finny, Pearson Publication, 9 th Edition, 2010.
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	L	H	H	H	H	L	H	-	-
CO2	M	M	M	H	M	M	L	H	L	-
CO3	L	H	L	L	L	H	L	H	L	-
CO4	M	H	L	M	H	H	L	H	-	-
CO5	M	M	M	M	H	H	L	H	-	-

H-High; M-Medium; L-Low

Core Paper – II	DIFFERENTIAL CALCULUS		
Course code	Total Hours	Contact Hours	Credits
25UMAC 02	4	4	4
	Course Objectives		
	The main objectives of this course are		
1	The basic skills of differentiation, successive differentiation, and their applications		
2	Basic knowledge on the notions of curvature, evolutes, involutes And polar co-ordinates and in solving related problems.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Find the nth derivative, form equations involving derivatives and apply Leibnitz formula.	K1 K2 K3 K4 K5 K6	
CLO 2	Find the partial derivative and total derivative coefficient.		
CLO 3	Determine maxima and minima of functions of two variables and to use the Lagrange’s method of undetermined multipliers.		
CLO 4	Find the envelope of a given family of curves.		
CLO 5	Find the evolutes and involutes and to find the radius of curvature using polar co- ordinates.		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Successive Differentiation: Introduction (Review of basic concepts) – The n^{th} derivative - Standard results -Fractional expressions - Trigonometrical transformation – Formation of equations involving derivatives - Leibnitz formula for the n^{th} derivative of a product. (Chapter 3: Sections1.1to 1.6 and 2.1, Related problems.)		
Unit-II	Partial Differentiation: Partial derivatives - Successive partial derivatives - Function of a function rule - Total differential coefficient - A special case - Implicit Functions. (Chapter 8: Sections1.1 to 1.5.)		
Unit-II	Partial Differentiation (Continued): Homogeneous functions - Partial		

I	derivatives of a function of two variables - Maxima and Minima of functions of two variables – Lagrange’s method of undetermined multipliers. (Chapter 8: Sections 1.6, 1.7 and Sections 4,5.)				
Unit-IV	Envelope: Method of finding the envelope - Another definition of envelope -Envelope of family of curves which are quadratic in the parameter. (Chapter 10: Sections 1.1 to 1.4.)				
Unit-V	Curvature: Definition of Curvature - Circle, Radius and Centre of Curvature - Cartesian formula for the radius of curvature - The coordinates of the centre of curvature - Evolutes and Involutives - Radius of Curvature in Polar Co-ordinates. (Chapter 10: Sections 2.1 to 2.6)				
Text Books	S. Narayanan and T.K. Manicavachagom Pillay, Calculus-Volume I,(2004),S. Viswananthan Printers Pvt. Ltd.				
Reference Books	1. H.Anton, I.Birens and S.Davis, Calculus,John Wiley and Sons, Inc., 2002. 2. G.B.Thomas and R.L.Finney, ,Pearson Education, 2010. 3. M.J.Strauss, G.L.Bradley and K. J.Smith, Calculus, 3 rd Ed., Dorling Kindersley(India) P.Ltd. (Pearson Education), Delhi, 2007. 4. R.Courant and F.John, Introduction to Calculus and Analysis(Volumes I & II), Springer - Verlag,New York, Inc., 1989. 5. T.Apostol,Calculus,Volumes I and II. 6. S.Goldberg, Calculus and mathematical analysis.				
Web Resources	https://nptel.ac.in				
Pedagogy					
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.					
<table border="1" style="width:100%; border-collapse: collapse;"> <tr> <td style="width:50%; text-align: center;">Course Designer (Name & Signature)</td> <td style="width:50%; text-align: center;">Verified by BoS Chairman</td> </tr> <tr> <td style="height: 40px;"></td> <td style="height: 40px;"></td> </tr> </table>		Course Designer (Name & Signature)	Verified by BoS Chairman		
Course Designer (Name & Signature)	Verified by BoS Chairman				

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	H	L	H	H	L	H	-	-
CO2	M	L	L	M	M	M	L	H	-	-
CO3	L	H	M	H	L	H	M	H	M	-

CO4	M	M	H	H	H	H	M	H	M	L
CO5	L	L	M	L	L	H	M	H	M	L

H-High; **M**-Medium; **L**-Low

Core Paper – III	ANALYTICAL GEOMETRY(Two & Three Dimensions)		
Course code	Total Hours	Contact Hours	Credits
25UMAC03	4	4	4
	Course Objectives		
	The main objectives of this course are		
1	Necessary skills to analyse characteristics and properties of two- and three-dimensional geometric shapes.		
2	To present mathematical arguments about geometric relationships.		
3	To solve real world problems on geometry and its applications.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Find pole, polar for conics, diameters, conjugate diameters for ellipse and hyperbola.	K1 K2 K3 K4 K5 K6	
CLO 2	Find the polar equations of straight line and circle, equations of chord, tangent and normal and to find the asymptotes of hyperbola.		
CLO 3	Explain in detail the system of Planes.		
CLO 4	Explain in detail the system of Straight lines.		
CLO 5	Explain in detail the system of Spheres.		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Ellipse - Parametric Representation - Pole and Polar - conormal points- Concyclic points and conjugate diameters - Properties of conjugate diameters - Equiconjugate diameters. (Book 1: Chapter 9)		
Unit-II	Polar coordinates: Distance between the points - Transformation of polar coordinates into cartesian coordinates - Area of triangle when the polar coordinates of the angular points are known - Equation of a straight line-parallel straight lines - perpendicular straight lines - Circle -Chord - Polar equation of a conic - Tracing of conic. (Book 2: Chapter 9 Section 1 to 9.1)		
Unit-III	System of Planes - Length of the perpendicular - Orthogonal projection. (Book 3: Chapter 2: Sections 2.5, 2.7, 2.9).		
Unit-IV	Representation of line-angle between a line and a plane - co-planar lines - shortest distance between two skew lines - length of the perpendicular - intersection of three planes. (Book 3: Chapter 3: Sections 3.1, 3.2, 3.4, 3.6, 3.7, 3.8) (Chapter 10: Sections 1.1 to 1.4.)		
Unit-V	Equation of a sphere -general equation - section of a sphere by a plane-equation of the circle - tangent plane - angle of intersection of two spheres - condition for the orthogonality - radical plane. (Book 3: Chapter 6: Sections 6.1, 6.2, 6.3, 6.4, 6.6, 6.7, 6.8)		

Text Books	1.Vittal P.R. and Malini V, Algebra, Analytical Geometry & Trigonometry, Margam Publications, India.2018. 2.Manicavachagom Pillay T.K. and Natarajan T, A Textbook of Analytical Geometry Part I-Two Dimensions, Divya Subramanian for Ananda Book Depot. 1996. 3.Shanti Narayan and Mittal P.K., Analytical Solid Geometry,S Chand Publishing, 2021.
Reference Books	1. S.L. Loney, Co-ordinate Geometry. 2. Robert J. T. Bell, Co-ordinate Geometry of Three Dimensions. 3. William F. Osgood and William C. Graustein, Plane and Solid Analytic Geometry, Macmillan Company, New York, 2016. 4. Calculus and Analytical Geometry, G.B.Thomas and R.L.Finny, Pearson Publication, 9 th Edition, 2010. 5. Robert C. Yates, Analytic Geometry with Calculus, Prentice Hall,Inc., New York, 1961. 6. Earl W. Swokowski and Jeffery A.Cole, Algebra and Trigonometry with Analytic Geometry, Twelfth Edition, Brooks/Cole, Cengage Learning, CA, USA, 2010. 7. William H. Mc Crea, Analytical Geometry of Three Dimensions, Dover Publications, Inc, New York, 2006. 8. John F. Randelph, Calculus and Analytic Geometry, Wadsworth Publishing Company, CA,USA, 1969. 9. Ralph Palmer Agnew, Analytic ,Mc Graw - Hill Book Company, nc.New York, 1962.
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	H	L	H	M	M	H	L	-
CO2	L	L	L	M	M	H	H	H	M	-
CO3	M	H	L	H	L	M	L	M	H	-

CO4	L	L	M	L	M	H	M	H	M	-
CO5	M	H	H	H	L	H	M	M	L	-

H-High; M-Medium; L-Low

Core Paper – IV	INTEGRAL CALCULUS		
Course code	Total Hours	Contact Hours	Credits
25UMAC04	4	4	4
	Course Objectives		
	The main objectives of this course are		
1	Knowledge on integration and its geometrical applications, double, triple integrals and improper integrals.		
2	Knowledge about Beta and Gamma functions and their applications.		
3	Skills to Determine Fourier series expansions.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Determine the integrals of algebraic, trigonometric and logarithmic functions and to find the reduction formulae	K1 K2 K3 K4 K5 K6	
CLO 2	Evaluate double and triple integrals and problems using change of order of integration		
CLO 3	Solve multiple integrals and to find the areas of curved surfaces and volumes of solids of revolution		
CLO 4	Explain beta and gamma functions and to use them in solving problems of integration		
CLO 5	Explain Geometric and Physical applications of integral calculus		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Reduction formulae - Types, integration of product of powers of algebraic and trigonometric functions , integration of product of powers of algebraic and logarithmic functions - Bernoulli’s formula. (Chapter 1: Sections 13 and 14)		
Unit-II	Multiple Integrals - definition of double integrals - evaluation of double integrals - double integrals in polar coordinates - Change of order of integration. (Chapter 5: Sections 1, 2.1, 2.2 and 3.1)		
Unit-III	Triple integrals - applications of multiple integrals - volumes of solids of revolution - areas of curved surfaces - change of variables -		

	Jacobian. (Chapter 5: Sections 4, 5.1, 5.2, 5.3, 6.1, 7 and Chapter 6: 1.1, 1.2)
Unit-IV	Beta and Gamma functions - infinite integral - definitions - recurrence formula of Gamma functions -properties of Beta and Gamma functions - relation between Beta and Gamma functions - Applications. (Chapter 7: Sections 2.1, 2.2, 2.3, 3, 4 and 6.)
Unit-V	Geometric Applications of Integration - Areas under plane curves: Cartesian coordinates - Area of a closed curve - Areas in polar coordinates -Trapezoidal rule - Simpson's rule and Physical Applications of Integral calculus - Centroid - Centre of mass of an arc - Centre of mass of a plane area - Centroid of a solid of revolution - Centroid of a surface of revolution . (Chapter2:Sections 1.1 to1.4 , 2.1,2.2and Chapter3: 1.1to 1.5 Simple Applications)
Text Books	Narayanan S and Manicavachagom Pillay T.K. Calculus -Volume II, (2006), S. Viswananthan Printers Pvt. Ltd.
Reference Books	1. H.Anton,I.Birens and S.Davis,Calculus, John Wiley and Sons, Inc., 2002. 2. G.B.Thomas and R.L.Finney, Calculus, Pearson Education,2 007. 3. D.Chatterjee, Integral Calculus and Differential Equations, Tata-Mc Graw Hill Publishing Company Ltd. 4. P.Dyke, An Introduction to Laplace Transforms and Fourier Series, Springer Undergraduate Mathematics Series, 2001 (second edition)
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	L	M	M	H	H	H	-	-
CO2	M	L	M	L	L	H	M	M	-	-

CO3	L	H	H	H	H	H	L	H	-	-
CO4	H	M	H	L	M	H	H	H	-	-
CO5	M	L	L	M	H	H	M	H	-	M

H-High; **M**-Medium; **L**-Low

SKILL ENHANCEMENT COURSE

NME Paper – I	MATHEMATICS FOR COMPETITIVE EXAMINATION–I (Non Major Elective)		
Course code	Total Hours	Contact Hours	Credits
25UMANM1	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Remembering the meaning of HCF and LCM of numbers.		
2	Understanding the concept of Decimal Fractions.		
3	Analyzing the concepts of Square Roots and Cube Roots		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Perform basic mathematics in Numbers.	K1 K2 K3 K4 K5 K6	
CLO 2	Understand Decimal Fractions and Simplification.		
CLO 3	Develop basic concept of Square Roots and Cube Roots and Average.		
CLO 4	Explain Problems on Numbers - Problems on Ages.		
CLO 5	Critique and evaluate quantitative arguments that utilize mathematics, statistical and quantitative informations.		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Numbers(Chapter– 1)		
Unit-II	H.C.F and L.C.M.of Numbers.(Chapter– 2)		
Unit-III	Decimal Fractions – Simplification. (Chapter– 3&4)		
Unit-IV	Square Roots and Cube Roots –Average. (Chapter– 5 & 6)		
Unit-V	Problems on Numbers(Chapter– 7)		
Text Books	R.S. Aggarwal, Quantitative Aptitude for Competitive Examinations, S.Chand co Ltd., 152. Anna Salai, Chennai, 2010.		
Reference Books	Quantitative Aptitude - by Abhijit Guha, Tata Mc Graw Hill Publishing Company Limited, New Delhi(2005)		
Web	https://nptel.ac.in		

Resources	
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	H	H	H	L	H	H	H	-	-
CO2	L	L	M	L	M	M	M	H	L	-
CO3	M	M	L	M	H	H	L	H	M	-
CO4	M	H	H	L	H	H	L	H	-	-
CO5	H	H	H	L	L	H	M	H	-	-

H-High; **M**-Medium; **L**-Low

Electiv e Paper – I	SEC I – FINANCIAL MATHEMATICS		
Course code	Total Hours	Contact Hours	Credits
25UMAS 01	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	To develop concepts of Geometric Brownian motion, Option pricing		
2	To create basic concepts of Probability theory		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	The basic concepts of Probability theory, The Central limit theorem.	K1 K2 K3 K4 K5 K6	
CLO 2	The concepts of Geometric Brownian motion, Option pricing		
CLO 3	The derivatives of Blackschole formula and its applications.		
CLO 4	The concept of call option on Dividend paying securities, estimating the volatility parameter.		
CLO 5	The limitations of Arbitrage pricing, the portfolio selection problem		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Probabilities and Events-Conditional Probability- -Random Variables and Expected Values-Covariance and Correlation-Continuous Random Variables-Normal Random Variables –Properties of Normal Random Variables-The Central Limit Theorem-Simple Problems. Section (1.1 - 1.4, 2.1 - 2.4)		
Unit-II	Geometric Brownian Motion-Geometric Brownian Motion as a limit of Simple ModelsBrownian Motion-Simple Problems- Interest Rates – Present Value Analysis- Rate of ReturnContinuously Varying Interest Rates-An example of option Pricing –Other example of Pricing via Arbitrage. Section (3.1 - 3.3, 4.1 - 4.4, 5.1, 5.2)		
Unit-II I	The Arbitrage Theorem-The Multi period Binomial Model-Proof of the Arbitrage Theorem-The Black Scholes Formula-Properties of the Black –Sholes Option Cost-Derivation of Black Scholes Formula- Simple Problems. Section (6.1 - 6.3, 7.2, 7.3, 7.5)		
Unit-IV	Additional results on options- Call option on Divided Paying Securities-Pricing American Put Options-Adding Jumps to Geometric Brownian Motion-Estimating the Volatility Parameter- Simple Problems. Section (8.2 - 8.5)		
Unit-V	Valuing by Expected Utility-Limitation of Arbitrage pricing-Valuing Investments by Expected Utility-The portfolio selection problem—Value at risk and conditional value at risk The capital assets pricing model-Mean Variances analysis of risk-Neutral priced Call optionsRates of return-Single Period and Geometric		

	Brownian Motion-- Simple Problems. Section (9.1 - 9.7)
Text Books	1. An Elementary Introduction to Mathematical Finance, 2nd Edition, Sheldon M. Ross, Cambridge University press, 2005
Reference Books	1. A First Course in Probability, S.M. Ross, Englewood cliffs Prentice Hall NJ, 2002 2. Option Market, J. Cox M. Rubinstein, Englewood cliffs Prentice Hall NJ, 1985
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	L	H	M	H	L	H	-	-
CO2	M	M	H	H	M	M	M	H	L	-
CO3	L	M	L	M	L	H	H	M	M	-
CO4	H	L	M	L	H	H	L	H	-	-
CO5	H	L	H	M	H	H	M	H	-	-

H-High; M-Medium; L-Low

Electiv e Paper – I	SEC II – OFFICE AUTOMATION (PRACTICAL)		
Course code	Total Hours	Contact Hours	Credits
25UMAS 02	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	To learn about basic commands of MS Word, MS Excel and MS Access.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Describe the history, objectives, and structure of the National Service Scheme.	K1 K2 K3 K4 K5 K6	
CLO 2	Identify social issues and structures within Indian society, including caste, gender, and regional disparities.		
CLO 3	Demonstrate awareness of the Indian Constitution and key principles of social justice.		
CLO 4	Participate meaningfully in NSS activities that promote national integration and community development.		
CLO 5	Reflect on the significance of values and civic responsibility in a democratic society		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
LIST OF PRAC TICAL S	MS Word 1. Preparation of word document (Typing, aligning, Font Style, Font Size, Text editing, colouring, Spacing, Margins) 2. Creating and Editing a table (Select no of rows, Select no of columns, row heading, column heading, column width, row width, row height, spacing text editing) 3. Formatting a table (insert rows/columns, delete rows/columns, cell merging / splitting, Cell alignment) 4. Preparation of letters using mail merge. 5. Demonstration of Find, Replace, Cut, Copy and paste texts in a word document. MS Excel 6. Creation of Charts, Graphs and Diagrams. 7. Calculation of Measures of central Tendency 8. Calculation of Standard Deviation. MS Power Point 9. Preparation of slides in power point. 10. Creation of Animation Pictures. MS Access 11. Creation of simple reports using MS Access. General 12. Exporting a given graph from Excel to word. 13. Sending an Email. 14. Downloading a document from internet.		

	15. Importing a picture from internet to word document.
Text Books	Andy Channelle, Beginning Open Office 3: From Novice to Professional, A Press series, Springer-Verlog, 2009.
Reference Books	1. 1. Perry M. Greg, Sams Teach Yourself Open Office.org All In One, Sams Publications, 2007.
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	L	H	M	H	L	H	-	-
CO2	M	M	H	H	M	M	M	H	L	-
CO3	L	M	L	M	L	H	H	M	M	-
CO4	H	L	M	L	H	H	L	H	-	-
CO5	H	L	H	M	H	H	M	H	-	-

H-High; M-Medium; L-Low

NME Paper – I	Quantitative Aptitude –I(Non Major Elective)		
Course code	Total Hours	Contact Hours	Credits
24UMANM1	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Remembering the meaning of HCF and LCM of numbers.		
2	Understanding the concept of Decimal Fractions.		
3	Analyzing the concepts of Square Roots and Cube Roots		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Perform basic mathematics in Numbers.	K1 K2 K3 K4 K5 K6	
CLO 2	Understand Decimal Fractions and Simplification.		
CLO 3	Develop basic concept of Square Roots and Cube Roots and Average.		
CLO 4	Explain Problems on Numbers - Problems on Ages.		
CLO 5	Critique and evaluate quantitative arguments that utilize mathematics, statistical and quantitative informations.		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Numbers(Chapter– 1)		
Unit-II	H.C.F and L.C.M.of Numbers.(Chapter– 2)		
Unit-III	Decimal Fractions – Simplification. (Chapter– 3&4)		
Unit-IV	Square Roots and Cube Roots –Average. (Chapter– 5 & 6)		
Unit-V	Problems on Numbers(Chapter– 7)		
Text Books	R.S. Aggarwal, Quantitative Aptitude for Competitive Examinations, S.Chand co Ltd., 152. Anna Salai, Chennai, 2010.		
Reference Books	Quantitative Aptitude - by Abhijit Guha, Tata Mc Graw Hill Publishing Company Limited, New Delhi(2005)		
Web Resources	https://nptel.ac.in		
Pedagogy			
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.			
Course Designer (Name & Signature)		Verified by BoS Chairman	

NME Paper – II	Quantitative Aptitude —II (Non Major Elective)		
Course code	Total Hours	Contact Hours	Credits
25UMANM2	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Understanding the concepts of chain rule		
2	Applying the concept of time and distance.		
3	Analyzing the problem on trains with solved examples.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Explain in detail about Problems on Ages.	K1 K2 K3 K4 K5, K6	
CLO 2	Explain Partnership and Chain Rule.		
CLO 3	Explain Time & Work.		
CLO 4	Explain Time& Distance and Problems on Trains.		
CLO 5	Explain Boats & Streams.		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Problems on Ages. (Chapter – 8)		
Unit-II	Chain Rule. (Chapter – 14)		
Unit-III	Time&Work. (Chapter – 15)		
Unit-IV	Time & Distance – Problems on Trains.(Chapter– 17&18)		
Unit-V	Boats & Streams. (Chapter – 19)		
Text Books	R.S. Aggarwal, Quantitative Aptitude for Competitative Examinations, S.Chand co Ltd., 152. Anna Salai, Chennai, 2010.		
Reference Books	Quantitative Aptitude -‘by Abhijit Guha, Tata Mc Graw Hill Publishing Company Limited, New Delhi(2005).		
Web Resources	https://nptel.ac.in		
Pedagogy			
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.			
Course Designer (Name & Signature)		Verified by BoS Chairman	

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	L	H	M	H	L	H	-	-
CO2	M	M	H	H	M	M	M	H	L	-
CO3	L	M	L	M	L	H	H	M	M	-
CO4	H	L	M	L	H	H	L	H	-	-
CO5	H	L	H	M	H	H	M	H	-	-

H-High; **M**-Medium; **L**-Low

ELECTIVE/ALLIED MATHEMATICS

Elective Paper – I	ALLIED MATHEMATICS-I: ALGEBRA AND CALCULUS(FORB.Sc PHYSICS/ B.Sc CHEMISTRY/B.Sc STATISTICS/ B. Sc., ELECTRONICS & COMMUNICATIONS)		
Course code	Total Hours	Contact Hours	Credits
25UMAG E1	6	6	4
	Course Objectives		
	The main objectives of this course are		
1	To learn the basic concepts and problem solving in Theory of equations.		
2	Develop the ability of solving the Integrals.		
3	Analyzing the problem on trains with solved examples.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Explain in detail about Imaginary roots, irrational roots and formation of equations and Descarte’s rule of sign.	K1 K2 K3 K4 K5 K6	
CLO 2	Explain Characteristic equation and roots of the matrix and Eigen values and Eigen vector of the matrix and Verification of Cayley Hamilton theorem.		
CLO 3	Explain Formula for Radius of curvature in Cartesian coordinates and Parametric coordinates and Polar coordinates		
CLO 4	Explain Formation of Partial Differential Equations by eliminating the arbitrary constant and arbitrary functions		
CLO 5	Explain Simple properties of definite Integrals and Bernoulli’s Formula and Integration by parts.		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Theory of Equations : Imaginary roots–Irrational roots–Formation of equations–Solutions of equations–Diminishing the roots of an equation & solutions – Removal of the second term of an equation & solutions –Descarte’s rule of sign–Problems only.(Chapter 6:Sections 4,9,10 & 11)		
Unit-II	Matrices: Definition of Characteristic equation of a matrix –Characteristic roots of a matrix -Eigen values and the Corresponding Eigen vectors of matrix–Cayley Hamilton theorem(Statement only)–Verifications Of Cayley Hamilton Theorem–Problems only.(Chapter5)		
Unit-III	Radius of Curvature: Formula of Radius of Curvature in Cartesian coordinates, Parametric coordinates and Polar coordinates (no proof for formulae) – Problems		

	only.(Chapter11)
Unit-IV	Partial Differential Equations Formation of Partial Differential Equations by eliminating the arbitrary constant and arbitrary functions – Lagrange’s Linear Partial Differential Equations– Problems only. (Chapter26)
Unit-V	Integration: Definite Integral: Simple properties of definite Integrals(Chap-15)–Bernoulli’s Formula–Integration by parts–Simple problems; Reduction formula for $\int_0^{\pi} \sin^n x dx$, $\int_0^{\pi} \cos^n x dx$, $\int_0^{\infty} e^{-x} dx$, $\int x^n e^{ax} dx$ simple problems.(Chapter16)
Text Books	Dr.P.R.Vittal, Allied Mathematics, Margham publication, Chennai–17, Reprint 2016.
Reference Books	1. S.G Venkatachalapathi, Allied Mathematics, Margham publication, Chennai– 17, Reprint 2011. 2. P.Kandasamy,K.Thilagavathy,Allied Mathematics VolumeI,S.Chand publication, July2012. 3. P.Kandasamy, K.Thilagavathy Allied Mathematics VolumeI I,S.Chand publication, December2010.
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	M	H	L	H	H	H	L	H	-	-
CO2	M	M	M	L	M	M	L	H	L	-
CO3	L	L	H	M	L	H	M	H	M	-
CO4	L	H	H	H	L	M	H	H	-	-
CO5	H	L	M	M	M	M	M	H	-	-

H-High; M-Medium; L-Low

Elective Paper – I	ALLIED MATHEMATICS-II: DIFFERENTIAL EQUATIONS AND LAPLACE TRANSFORMS (FOR B.Sc., PHYSICS/B. Sc., CHEMISTRY/B.Sc., STATISTICS/ B. Sc., ELECTRONICS&COMMUNICATIONS)		
Course code	Total Hours	Contact Hours	Credits
25UMAGE2	6	6	4
	Course Objectives		
	The main objectives of this course are		
1	Develop the basic concepts of Maxima and Minima of two variables and Numerical methods problems.		
2	To learn the second order differential equation with constant coefficients.		
3	To learn the basic concepts of Laplace Transforms, Inverse Laplace Transforms &Applications.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Explain Jacobian of two variables and three variables and Maxima and Minima functions of two variables.	K1 K2 K3 K4 K5 K6	
CLO 2	Explain Finite difference and Higher differences and Construction of differentiable and Newton’s Forward Backward difference formula and Lagrange’s Interpolation formula.		
CLO 3	Explain Second Order Differential Equation with constant coefficients and Particular Integral		
CLO 4	Explain definition of Laplace Transforms and standard formula and linearity property and shifting property and Change of Scale property and Laplace Transforms of derivatives		
CLO 5	Explain standard formula and elementary theorems and Applications to solutions of second order differential equations with constant coefficients.		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		

Unit-I	Jacobian and Maxima & minima: Jacobian of two variables and three variables – Maxima and Minima functions of two variables – Problems only. (Chapter 9: Sections 3 &4)
Unit-II	Finite Differences: Finite difference – Higher differences – Construction of differentiable–Interpolation of missing value–Newton’s Forward and Newton’s Backward difference formula(no proof)–Lagrange’s Interpolation formula (no proof)-simple problems only. (Chapter 7)
Unit-III	Second Order Differential Equations: Second Order Differential Equation with constant coefficients - Complementary function – Particular Integral and Solution of the type: e^{ax} , x^n , $\cos ax$ (or) $\sin ax$, $e^{as}x^{bs}$, $e^{as}\sin bx$, $e^{as}\cos bx$ – Problems only. (Chapter 23)
Unit-IV	Laplace Transforms: Definition of Laplace Transforms –Standard formula–Linearity property – shifting property – Change of Scale property – Laplace Transforms of derivatives– Problems. (Chapter 27)
Unit-V	Inverse Laplace Transforms: Standard formula- Elementary theorems (no proof) – Applications to solutions of second order differential equations with constant coefficients – simple problems. (Chapter 27)
Text Books	Dr.P.R.Vittal, Allied Mathematics, Margham publication, Chennai–17, Reprint 2016
Reference Books	1. S.G Venkatachalapathi, Allied Mathematics, Margham publication, Chennai– 17, Reprint 2011 2. P.Kandasamy, K.Thilagavathy, Allied Mathematics VolumeI, S.Chand publication, July 2012 P.Kandasamy, K.Thilagavathy Allied Mathematics VolumeII, S.Chand publication, December 2010
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs	POs

	1	2	3	4	5	1	2	3	4	5
CO1	M	M	H	H	H	H	M	H	-	-
CO2	H	L	L	L	H	M	L	H	L	-
CO3	L	H	M	M	M	H	H	H	M	-
CO4	L	L	H	H	M	H	L	H	-	-
CO5	M	M	L	M	L	H	M	H	-	-

H-High; M-Medium; L-Low

Elective Paper – I	ALLIED MATHEMATICS – PRACTICAL (FOR B.Sc., PHYSICS/B. Sc., CHEMISTRY/B.Sc., STATISTICS/ B. Sc., ELECTRONICS & COMMUNICATIONS) ALLIED MATHEMATICS PRACTICAL		
Course code	Total Hours	Contact Hours	Credits
25UMAGE P01	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Acquire knowledge about Matrices and Cayley – Hamilton Theorem		
2	Understand the concepts of differentiation and Vector point function.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Explain in detail about Rank of Matrix and Characteristics equation of a Matrix and Cayley Hamilton Theorem and Problems to verify Cayley Hamilton .	K1 K2 K3 K4 K5 K6	
CLO 2	Explain Leibnitz formula for nth derivative		
CLO 3	Explain Euler’s theorem on homogeneous function and Problems to verify Euler’s Theorem and Partial derivative.		
CLO 4	Explain Scalar and Vector point functions and Gradient of scalar point functions.		
CLO 5	Explain Divergence and Curl of vector point functions and Solinoidal vector and Irrotational vector		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Matrices: Rank of Matrix – Problems upto (3x3) Matrix – Characteristics equation of a Matrix – Cayley Hamilton Theorem (statement only) –Problems		

	to verify Cayley Hamilton Theorem. (Chapter5)
Unit-II	Leibnitz formula for n^{th} derivative: Leibnitz formula (without proof) for n^{th} derivative – Problems.(Page no:8.23 to 8.39 of the Text book)(Chapter 8)
Unit-III	Partial Differentiation : Euler’s theorem on homogeneous function(without proof) – Problems to verify Euler’s Theorem – Partial derivative – problems (Page no. 9.1 to 9.13 and 9.18 to 9.27 of the Text Book)(Chapter 9)
Unit-IV	Vector Differentiation : Scalar and Vector point functions – Gradient of scalar point functions – Problems only.(Chapter 28)
Unit-V	Divergence and Curl of Vector point functions: Divergence and Curl of vector point functions – Solenoidal vector – Irrotational vector – Problems only.(Chapter 28)
Text Books	Dr.P.R.Vittal, Allied Mathematics, Margham publication, Chennai-17, Reprint 2016
Reference Books	1. S.G Venkatachalapathi, Allied Mathematics, Margham publication, Chennai – 17, Reprint 2011. 2. P.Kandasamy, K.Thilagavathy, Allied Mathematics Volume I, S.Chand publication, July 2012. P.Kandasamy, K.Thilagavathy Allied Mathematics Volume II, S.Chand publication, December 2010.
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature) Verified by BoS Chairman	

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	H	H	H	H	H	L	H	-	-
CO2	H	H	M	M	M	M	M	H	M	-
CO3	H	H	L	L	L	H	L	H	L	-

CO4	H	M	L	H	H	H	M	H	-	-
CO5	H	M	M	M	H	H	L	H	-	-

H-High; **M**-Medium; **L**-Low

Elective Paper	DISCRETE MATHEMATICS – I (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAG E3	6	6	3
	Course Objectives		
	The main objectives of this course are		
1	Mathematical Logic		
2	Truth Table		
3	Relations and Ordering		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Define Mathematical Logics and few examples	K1 K2 K3 K4 K5 K6	
CLO 2	Define Normal Forms and The theory of inference for the statement calculus		
CLO 3	Describe The Predicate Calculus and Inference theory of the predicate calculus		
CLO 4	Define Some Basic Set Identities, and Cartesian products		
CLO 5	Describe Relation and ordering and Functions		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Mathematical Logic: Statement and Notation - Connectives – Negation – Conjunction – Disjunction-Statement Formulas and Truth Tables – Conditional and Biconditional – Well-formed Formulas – Tautologies Chapter: 1 (Section: 1.1, 1.2(1.2.1 to 1.2.8))		
Unit-II	Normal Forms: Disjunctive Normal Forms – Conjunctive Normal Forms –Principal Disjunctive Normal Forms –Principal Conjunctive Normal Forms –Ordering and Uniqueness of Normal Forms The theory of inference for the statement calculus: Validity Using Truth Tables -Rules of Inference – Consistency of Premises and Indirect Method of Proof. Chapter-1 (section: 1.3, 1.3.1 to 1.3.5 & 1.4, 1.4.1 to 1.4.3)		
Unit-III	The Predicate Calculus: Predicates – The Statement Function, Variables, and Quantifiers – Predicate Formulas – Free and Bound Variables – The Universe of Discourse. Inference theory of the predicate calculus: Valid Formulas and Equivalences – Some Valid Formulas Over Finite Universes – Special Valid Formulas Involving Quantifiers – Theory of Inference for the Predicate Calculus – Formulas Involving More Than One Quantifiers. Chapter-1 (section: 1.5, 1.5.1 to 1.5.5 & 1.6.1 to 1.6.5)		
Unit-IV	Set Theory: Notation – Inclusion and Equality of Sets – The Power Set – Some Operations on Sets –Venn Diagrams – Some Basic Set		

	Identities – The Principle of Specification – Ordered Pairs and n-tuples – Cartesian Products. Chapter-2(section: 2.1.1 to 2.1.9)
Unit-V	Relation and ordering: Relations – Properties of Binary Relations in a Set – Relation Matrix and the Graph of a Relation – Partition and Covering of a Set – Functions: Definition and Introduction – Composition of Function – Inverse Function – Binary and n-ary Operations – Characteristic Function of a Set – Hashing Functions-Peano Axioms and Mathematical Introduction – Cardinality. Chapter- 2 (section: 2.3.1 to 2.3.4 & 2.4.1 to 2.5.2)
Text Books	1. Discrete mathematics structures with application to computer science – J.P.Tremblay and R. Manohar
Reference Books	1. Discrete Mathematics – Dr.S.P.Rajagopalan and Dr.R.Sattanathan 2. Discrete Mathematics – Dr.G.Balaji 3. Discrete Mathematics and its applications – Kenneth.H.Rosen.
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	H	H	H	H	H	L	H	-	-
CO2	H	H	M	M	M	M	L	H	L	-
CO3	H	H	L	L	L	H	M	H	M	-
CO4	H	M	L	H	H	M	H	H	-	-
CO5	H	M	M	M	H	M	M	H	-	-

H-High; M-Medium; L-Low

Elective Paper	DISCRETE MATHEMATICS - II (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAG E4	6	6	3
	Course Objectives		
	The main objectives of this course are		
1	Mathematical Logic		
2	Truth Table		
3	Relations and Ordering		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Define Mathematical Logics and few examples	K1 K2 K3 K4 K5 K6	
CLO 2	Define Groups and The application of the residue arithmetic to computers		
CLO 3	Define Latex and Boolean algebra and problems		
CLO 4	Define Boolean functions and examples		
CLO 5	Define graph theory and some basic definitions		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Algebraic Systems: Examples and General Properties-Definition and Examples-Some Simple Algebraic Systems and General Properties. Semigroups and Monoids: Definitions and ExamplesHomomorphism of Semigroups and Monoids-Sub semigroups and Sub monoids Grammars and languages: Discuss of Grammars-Formal definition of a Language-Notion of Syntax Analysis (Chapter-3: Sections 3.1 to 3.3)		
Unit-II	Groups: Definitions and Examples-Subgroups and homomorphismsCosets and Language’s Theorem-Normal Subgroups-Algebraic systems with Two Binary operations-The application of the residue arithmetic to computers: Introduction to number system-residue arithmetic. (Chapter 3: Sections 3.5(3.5.1 - 3.6.2)		
Unit-III	Latex and Boolean algebra: Lattices as partially ordered sets-definition and examples-some properties of lattices-lattices as algebraic system-sublattices, Direct product, and homomorphism-some special lattices -Boolean algebra definition and examples-subalgebra, direct product, and homomorphism. (Chapter 4: Sections 4.1.1 to 4.2.2)		
Unit-IV	Boolean function-Boolean forms and free Boolean algebras-values of Boolean expressions and Boolean functions Representation and minimization of Boolean functions: representation of Boolean functions-minimization of Boolean functions(Chapter 4: Sections		

	4.3.1 to 4.4.2)
Unit-V	Graph theory: Basic concepts of graph theory-basic definitions-paths, reachability and connectedness-matrix representation of graphs-trees-storage representation and manipulation of graphs-Trees: their representation and operations-List: structures and graphs (Chapter 5: Sections 5.1.1 to 5.2.2)
Text Books	1. Discrete mathematics structures with application to computer science – J.P. Tremblay and R. Manohar
Reference Books	1. Discrete Mathematics – Dr.S.P. Rajagopalan and Dr.R. Sattanathan 2. Discrete Mathematics – Dr.G.Balaji 3. Discrete Mathematics and its applications – Kenneth.H.Rosen
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	M	M	H	H	H	H	L	H	-	-
CO2	H	L	L	L	H	M	L	H	L	-
CO3	L	H	M	M	M	H	M	H	M	-
CO4	L	L	H	H	M	M	H	H	-	-
CO5	M	M	L	M	L	M	M	H	-	-

H-High; M-Medium; L-Low

Elective Paper	NUMERICAL METHODS (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAG E5	6	6	3
	Course Objectives		
	The main objectives of this course are		
1	Numerical methods is a mathematical tool designed to solve numerical problems.		
2	It is the study of numerical methods that attempt at finding approximate solutions of problems rather than the exact ones.		
3	Apply Numerical differentiation and Numerical integration.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Solve Algebraic methods and problems	K1 K2 K3 K4 K5 K6	
CLO 2	Define Interpolation with equal intervals and problems		
CLO 3	Define Interpolation with unequal intervals and problems		
CLO 4	Define Numerical Differentiation and Integration, problems		
CLO 5	Define Solution of Simultaneous linear Algebraic equations and problems		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Solution of Algebraic Introduction -The Bisection Method - The Iteration method – The method of false position - Newton Raphson Method -Generalized Newton's Method (Chapter 2: Sections 2.1 to 2.5)		
Unit-II	Interpolation with equal intervals Finite Difference - Forward Differences -Backward Differences - Central Differences - symbolic relations and separation of symbols - Newton's formulae for interpolation - Central Differences interpolation formulae - Gauss Central difference formulae - Stirling Formulae (Chapter 3: Sections 3.3(3.3.1 - 3.3.4), 3.6, 3.7(3.7.1 - 3.7.2))		
Unit-III	Interpolation with unequal intervals Lagrange's Interpolation Formulae - Divided differences - Divided differences table - Newton's Divided Difference formulae - Inverse Interpolation. (Problems only) (Chapter 3: Sections 3.9.1, 3.11.1, 3.12)		
Unit-IV	Numerical Differentiation and Integration Numerical Differences - Maximum and minimum values of Tabulated function - Numerical Integration - Trapezoidal Rule - Simphson 1/3 Rule - Simphson 3/8 Rule (Chapter 5: Sections 5.2, 5.3, 5.4(5.4.1 - 5.4.3))		

Unit-V	Solution of Simultaneous linear Algebraic equations Direct method - Gauss elimination Method - Gauss Jordan Method - Modification of Gauss Method to compute the inverse - Method of Factorization - Iterative Methods -Gauss Jacobi method - Gauss seidel Method. (Problems only) (Chapter 6: Sections 6.3(6.3.2 - 6.3.4), 6.4)	
Text Books	1. S.S. Sastry - Introductory methods of numerical Analysis 3rd Edition, Prentice Hall of India Private Ltd., New Delhi.	
Reference Books	1. P. Kandasamy, K. Thilagavathy, K. Gunavathy -Numerical Methods, Third Revised Edition, S.Chand&Company Ltd., Ram Nagar, New Delhi.	
Web Resources	https://nptel.ac.in	
Pedagogy	Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)		Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	H	H	H	H	H	L	H	-	-
CO2	H	H	M	M	M	M	L	H	L	-
CO3	H	H	L	L	L	H	M	H	M	-
CO4	H	M	L	H	H	M	H	H	-	-
CO5	H	M	M	M	H	M	M	H	-	-

H-High; M-Medium; L-Low

Elective Paper	OPTIMIZATION TECHNIQUES (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAG E6	6	6	3
	Course Objectives		
	The main objectives of this course are		
1	To know the concepts of Mathematical formulation and solving L.P.P		
2	To find the solutions of Transportation and Assignment models.		
3	To teach the techniques for converting the real life problems as Mathematical problems and solving them.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Formulate and solve real life problems through L.P.P.	K1 K2 K3 K4 K5 K6	
CLO 2	Compute the optimum Transportation schedule.		
CLO 3	Find the optimum Assignment model.		
CLO 4	Solve Sequencing problems.		
CLO 5	Use the techniques for planning and scheduling of projects.		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Linear Programming Formulation and Graphical Method: Introduction - Requirements for employing LPP technique – Mathematical Formulation of L.P.P. - Basic assumptions - Graphical method of the Solution of a L.P.P. – Some more cases – Advantage of Linear Programming – Limitations of Linear Programming. Chapter 2 (Sections 2.1 – 2.8)		
Unit-II	Transportation Model: Introduction - Mathematical formulation of a transportation problem - Methods for finding initial basic feasible solution – Transportation algorithm or MODI method – Degeneracy in Transportation problems – Unbalanced Transportation Problems – Maximization case in Transportation problems. Chapter 7 (Sections 7.1 – 7.5)		
Unit-III	Assignment Problem: Introduction – Mathematical formulation of an Assignment Problem –Difference between the Transportation Problem and Assignment Problem – Assignment Algorithm or Hungarian Method – Unbalanced Assignment Models – Maximization case in Assignment Problems. Chapter 8 (Sections 8.1 – 8.2, 8.4 - 8.7)		
Unit-IV	Sequencing Problems: Introduction – Assumptions of solving a sequencing Problem - Definition - Procedure for finding Optimum Sequence (n jobs on 2 machines) – Processing n jobs on three		

	machines – Processing n jobs on m machines. Chapter 14 (Sections 14.1 – 14.6).
Unit-V	Scheduling by PERT and CPM: Introduction – Basic Terminologies – Rules for constructing a project network – Network computations – Floats – Programme Evaluation Review Technique (PERT) – Basic differences between PERT and CPM. Chapter 15 (Sections 15.1 – 15.7)
Text Books	1. Sundaresan,V., Ganapathy Subramanian, K.S. and Ganesan, K. Resource Management Techniques. [Seventh Edition]. AR Publication, Chennai.2013
Reference Books	1. Kantiswarup., Gupta, P.K. and Man Mohan. Operations Research.[Seventeenth Edition]. Sultan Chand and Sons, New Delhi.2020. 2. Gupta, P.K. and Hira,D.S. Operations Research. [Eighth Edition]. Sulthan Chand and Company, New Delhi .2020. 3. Kalavathy.S. Operations Research[Fourth Edition],Vikas Publishing House,Chennai.2012.
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	H	M	L	H	L	H	-	-
CO2	L	H	H	M	L	M	L	H	L	-
CO3	M	L	M	L	H	H	M	H	M	-
CO4	H	H	M	L	H	M	H	H	-	-
CO5	M	M	L	H	M	M	M	H	-	-

H-High; M-Medium; L-Low

Elective Paper	INTRODUCTION TO LINEAR ALGEBRA (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAG E7	6	6	3
	Course Objectives		
	The main objectives of this course are		
1	Develop the ability of solving the Partial fraction, Binomial Series Exponential series and Logarithms Series		
2	Acquire knowledge about Matrices and Cayley – Hamilton Theorem.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Define Partial Fraction and Binomial Series and examples	K1 K2 K3 K4 K5 K6	
CLO 2	Define Exponential Series and Logarithms Series and examples		
CLO 3	Define matrix and simple problems		
CLO 4	Define Rank of matrix and problems		
CLO 5	Describe Cayley Hamiltan Theorem		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Partial Fraction and Binomial Series: Partial Fraction-Resolution into partial fraction-Binomial theorem for a positive integral index-Binomial theorem for a rational indexSimple problems. Chapter-1 and 2		
Unit-II	Exponential Series and Logarithms Series: Exponential series-Standard result for exponential series-Logarithms Series-Simple problems. Chapter-3 and 4		
Unit-III	Matrices: Introduction- Type of matrix-Matrix Operations-Transpose of a matrix-Determinant of a matrix-Inverse of a matrix-symmetric and skew symmetric-Conjugate of a matrix-Hermitian and skew Hermitian matrix-Simple problems Chapter-5 (Page No:5.1 to 5.17)		
Unit-IV	Rank of a Matrix: Orthogonal and Unitary matrix – Rank of a matrix- Test tor consistency of linear equation-Condition for consistency Chapter-5 (Page No:5.18 to 5.49)		
Unit-V	Cayley Hamilton Theorem: Definition of Characteristic equation of a matrix –Characteristic roots of a matrix - Eigen values and the Corresponding Eigen vectors of matrix– Cayley Hamilton theorem (Statement only) – Verifications of Cayley Hamilton Theorem – Problems only. (Chapter 5) (Page No:5.50- 5.74)		
Text Books	Dr.P.R. Vittal, Allied Mathematics ,Margham publication, Chennai– 17, Reprint 2016.		
Reference	1. S.G Venkatachalapathi, Allied Mathematics, Margham publication,		

Books	Chennai – 17, Reprint 2011 2. P. Kandasamy, K.Thilagavathy Allied Mathematics Volume I, S.Chand publication, July2012. 3. P. Kandasamy, K.Thilagavathy Allied Mathematics Volume II, S. Chand publication, December 2010.
Web Resources	https://nptel.ac.in
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	H	M	L	H	L	H	-	-
CO2	L	H	H	M	L	M	L	H	L	-
CO3	M	L	M	L	H	H	M	H	M	-
CO4	H	H	M	L	H	M	H	H	-	-
CO5	M	M	L	H	M	M	M	H	-	-

H-High; M-Medium; L-Low

Elective Paper	GRAPH THEORY AND ITS APPLICATION (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAG E8	6	6	3
	Course Objectives		
	The main objectives of this course are		
1	Graphs and subgraphs		
2	Walks, Trails and Paths		
3	Applications		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Define Graphs, Subgraphs and Operation on Graphs.	K1 K2 K3 K4 K5 K6	
CLO 2	Define Walk, Trails and Paths.		
CLO 3	Define Eulerian Graphs and Hamiltonian graphs. Explain the concept of Konigsberg Bridge problem and Fleury’s Algorithms.		
CLO 4	Explain Characterization of Trees and Theorems.		
CLO 5	Explain Applications of Connector problem and shortest path problem.		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Introduction – Definition – Examples – Degrees – Definition – Theorem 1, 2 – Problems – Subgraphs – Definition – Theorems – Operations on graphs – Definition theorem-1 – Problems. Chapter 2 (Sections 2.1 to 2.3, 2.9).		
Unit-II	Introduction – Walks, Trails and Paths – Definitions Theorem – 1, 2, 3 – Connectedness and components – Definitions – Theorems – Definition – Distance – Theorems – Cut point – Bridge – Blocks – Connectivity. Chapter 4 (Sections 4.1 to 4.4).		
Unit-III	Introduction – Eulerian Graphs – Definition – Lemmas – Theorem – Konigsberg Bridge problem – Fleury’s Algorithms – Hamiltonian graphs – Definitions – Theorems – Lemma – Closure – Theorems. Chapter 5 0(Sections 5.1, 5.2).		
Unit-IV	Introduction – Characterization of Trees – Theorems – Centre of a tree – Definition – Theorem. Chapter 6 (Sections 6.1, 6.2).		
Unit-V	Introduction - Applications – Connector problem – shortest path problem – Transformation and kinematic Graph. Chapter 11 (Sections 11.1 to 11.3)		
Text Books	S. Arumugam, S. Ramachandran, Invitation to graph theory, Scitech Publications, Chennai, 2001.		
Reference Books	1. John clark and Derek Allan Holton, A first book at graph theory, Allied publishes.		

	2. S. Kumaravelu and SusheelaKumaravelu, Graph theory, Publishers Authors C/O.182, Childambara Nagar, Nagarkoil – 629 002.	
Web Resources	https://nptel.ac.in	
Pedagogy		
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.		
Course Designer (Name & Signature)		Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	L	H	H	H	H	L	H	-	-
CO2	M	M	M	H	M	M	L	H	L	-
CO3	L	H	L	L	L	H	M	H	M	-
CO4	M	H	L	M	H	M	H	H	-	-
CO5	M	M	M	M	H	M	M	H	-	-

H-High; M-Medium; L-Low

Elective Paper	NUMERICAL METHODS-I (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAG E9	6	6	3
	Course Objectives		
	The main objectives of this course are		
1	Numerical methods is a mathematical tool designed to solve		
2	Numerical problems.		
3	It is the study of numerical methods that attempt at finding.		
4	Approximate solutions of problems rather than the exact ones.		
5	Apply Numerical differentiation and Numerical integration.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Define Algebraic methods and problems	K1 K2 K3 K4 K5 K6	
CLO 2	Define Newtons methods and Root squaring methods and problems		
CLO 3	Define finite differences and problems		
CLO 4	Define Interpolation methods and problems		
CLO 5	Define divided differences and inverse interpolation and problems		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	The Bisection Method - The Iteration method - The method of false position - Newton Raphson Method (Chapter 2: Sections 2.1 to 2.5)		
Unit-II	Generalized Newton's Method - Ramanujan's Method-The Secant Method - Muller's Method-Graeffe’s Root squaring Method (Chapter 2: Sections 2.6 to 2.9))		
Unit-III	Finite Difference - Forward Differences -Backward Differences - Central Differences - symbolic relations and separation of symbols-Detection of Errors by Use of Difference Tables. (Chapter 3: Sections 3.3(3.3.1 - 3.3.4),3.4		
Unit-IV	Differences of Polynomial- Newton's formulae for interpolation – Central Differences interpolation formulae - Gauss Central difference formulae - Stirling Formulae - Bessel's Formulae -Everett's formulae (Problems only). (Chapter 3: Sections 3.5,3.6.3.7(3.7.1 - 3.7.4))		
Unit-V	Lagrange's Interpolation Formulae – Divided differences - Divided differences table - Newton's Divided Difference formulae - Inverse Interpolation. (Problems only) (Chapter 3: Sections 3.9.1, 3.11.1, 3.12)		
Text Books	S.S. Sastry - Introductory methods of numerical Analysis 3rd Edition, Prentice Hall of India Private Ltd., New Delhi.		
Reference	P. Kandasamy, K. Thilagavathy, K. Gunavathy - Numerical Methods,		

Books	Third Revised Edition, S.Chand& Companyy Ltd., Ram Nagar, New Delhi.	
Web Resources	https://nptel.ac.in	
Pedagogy		
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.		
Course Designer (Name & Signature)		Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	H	M	L	H	L	H	-	-
CO2	L	H	H	M	L	M	L	H	L	-
CO3	M	L	M	L	H	H	M	H	M	-
CO4	H	H	M	L	H	M	H	H	-	-
CO5	M	M	L	H	M	M	M	H	-	-

H-High; **M**-Medium; **L**-Low

Elective Paper	NUMERICAL METHODS-II (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAG E10	6	6	3
	Course Objectives		
	The main objectives of this course are		
1	Numerical methods is a mathematical tool designed to solve		
2	Numerical problems.		
3	It is the study of numerical methods that attempt at finding.		
4	Approximate solutions of problems rather than the exact ones.		
5	Apply Numerical differentiation and Numerical integration.		
CLO numbers	Expected Course Learning Outcome: On the successful completion of the course, student will be able to:		
CLO 1	Define Numerical differentiation and problems	K1 K2 K3 K4 K5 K6	
CLO 2	Define Numerical Integration and problems		
CLO 3	Define direct methods and number of arithmetic operations related problems		
CLO 4	Define Method of factorization and problems		
CLO 5	Define solution by Taylor's Series and problems		
	K1 - Application; K2 -Foundational; K3 - Integration; K4 – Human Dimension; K5 – Caring K6 – Learning How to Learn		
	Course content		
Unit-I	Introduction- Numerical Differentiation - Maximum and minimum values of Tabulated function. (Chapter 5: Sections 5.1, 5.2,5.3)		
Unit-II	Numerical Integration-Trapezoidal Rule - Simphson 1/3 Rule - Simphson 3/8 Rule -Boole's and Weddle's rule. (Problems only) (Chapter 5: Sections 5.4(5.4.1 - 5.4.4))		
Unit-III	Direct method –Matrix Inversion Method-Gauss elimination Method – Gauss Jordan Method - Modification of Gauss Method to compute the inverse -Number of Arithmetic Operations-LU Decomposition-LU Decomposition from Gauss Elimination (Chapter 6: Sections 6.3(6.3.1 - 6.3.7))		
Unit-IV	Method of Factorization - Iterative Methods -Gauss Jacobi method - Gauss seidel Method. (Problems only) (Chapter 6: Sections 6.4)		
Unit-V	Solution by Taylor’s Series-Picard’s Method of Successive Approximations-Eluler’s Method-Runge-Kutta Method. (Chapter 7: Sections 7.2 to 7.5)		
Text Books	S.S. Sastry - Introductory methods of numerical Analysis 3rd Edition, Prentice Hall of India Private Ltd., New Delhi.		
Reference Books	P. Kandasamy, K. Thilagavathy, K. Gunavathy - Numerical Methods, Third Revised Edition, S.Chand& Company Ltd., Ram Nagar, New Delhi.		
Web	https://nptel.ac.in		

Resources	
Pedagogy	
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.	
Course Designer (Name & Signature)	Verified by BoS Chairman

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	M	H	M	L	H	L	H	-	-
CO2	L	H	H	M	L	M	L	H	L	-
CO3	M	L	M	L	H	H	M	H	M	-
CO4	H	H	M	L	H	M	H	H	-	-
CO5	M	M	L	H	M	M	M	H	-	-

H-High; **M**-Medium; **L**-Low

Elective Paper	DISCRETE MATHEMATICS – PRACTICAL (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAGEP 02	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Acquire knowledge about Bisection Method and Iteration Method.		
2	Understand the concepts of Gauss Jacobi Method and Gauss Seidel Method.		
	Course content		
Unit-I	Negation – Conjunction – Disjunction – Tautologies.		
Unit-II	Conditional – Biconditional.		
Unit-III	Boolean Algebra – Boolean functions.		
Unit-IV	Characteristic equation of Matrix – Characteristic roots.		
Unit-V	Cayley Hamilton Theorem – Rank of Matrix		
Text Books	1. Discrete Mathematics Structures with application to computer science – J.P. Tremblay and R. Manohar. 2. Discrete Mathematics – B.S. Vatssa, Suchi Vatssa.		
Reference Books	1. Discrete Mathematics – Dr.S.P.Rajagopalan and Dr.R.Sattanathan 2. Discrete Mathematics – Dr.G.Balaji 3. Discrete Mathematics and its applications – Kenneth.H.Rosen		
Web Resources	https://nptel.ac.in		
Pedagogy			
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.			
Course Designer (Name & Signature)		Verified by BoS Chairman	

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	H	H	H	L	H	L	H	-	-
CO2	L	L	M	L	M	M	M	H	M	-
CO3	M	M	L	M	H	H	L	H	L	-
CO4	M	H	H	L	H	H	M	H	-	-

CO5	H	H	H	L	L	H	L	H	-	-
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H-High; M-Medium; L-Low

Elective Paper	NUMERICAL METHODS – PRACTICAL ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAGE P03	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Acquire knowledge about Bisection Method and Iteration Method.		
2	Understand the concepts of Gauss Jacobi Method and Gauss Seidel Method.		
	Course content		
Unit-I	The Bisection Method – The Iteration Method.		
Unit-II	Newton's Interpolation Formula for Forward and Backward difference.		
Unit-III	Lagrange's Interpolation Formula.		
Unit-IV	Gauss elimination Method – Gauss Jordan Method.		
Unit-V	Gauss Jacobi Method – Gauss seidel Method		
Text Books	S.S. Sastry – Introductory Methods of numerical Analysis 3rd Edition, Prentice Hall of India Private Ltd., New Delhi.		
Reference Books	P. Kandasamy, K. Thilagavathy, K. Gunavathy – Numerical Methods, Third Revised Edition, S. Chand & Company Ltd., Ram Nagar, New Delhi.		
Web Resources	https://nptel.ac.in		
Pedagogy	Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.		
Course Designer (Name & Signature)		Verified by BoS Chairman	

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	H	H	H	L	H	L	H	-	-
CO2	L	L	M	L	M	M	M	H	M	-

CO3	M	M	L	M	H	H	L	H	L	-
CO4	M	H	H	L	H	H	M	H	-	-
CO5	H	H	H	L	L	H	L	H	-	-

H-High; M-Medium; L-Low

Elective Paper	OPTIMIZATION TECHNIQUES – PRACTICAL (FOR ALL COMPUTER SCIENCE DEPARTMENTS)		
	ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAGE P04	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Acquire knowledge about LPP and Graphical Method.		
2	Understand the concepts of Rules for constructing a project network.		
	Course content		
Unit-I	Linear Programming Problem – Graphical Method.		
Unit-II	Transportation Problem – Finding initial basic feasible solution only by using North-West Corner Rule, Least Cost Method – Vogel’s Approximation Method.		
Unit-III	Assignment Problem – Finding optimal solution by using Hungarian Method.		
Unit-IV	Sequencing Problem – N jobs to be operated on Two Machines.		
Unit-V	Network - Rules for constructing a project network – Floats.		
Text Books	1. Sundaresan. V., Ganapathy Subramanian K.S. and Ganesan. K, Resource Management Techniques. [Seventh Edition]. AR Publication, Chennai.2013.		
Reference Books	1. Kantiswarup., Gupta, P.K. and Man Mohan. Operations Research [Seventeenth Edition]. Sultan Chand and Sons, New Delhi.2020. 2. Gupta, P.K. and Hira,D.S. Operations Research [Eighth Edition]. Sultan Chand and Sons, New Delhi.2020. 3. Kalavathy.S. Operations Research [Fourth Edition], vikas publishing House, Chennai.2012.		
Web Resources	https://nptel.ac.in		
Pedagogy	Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.		
Course Designer (Name & Signature)		Verified by BoS Chairman	

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	H	H	H	H	H	L	H	-	-
CO2	H	H	M	M	M	M	M	H	M	-
CO3	H	H	L	L	L	H	L	H	L	-
CO4	H	M	L	H	H	H	M	H	-	-
CO5	H	M	M	M	H	H	L	H	-	-

H-High; M-Medium; L-Low

H-High, M-Medium, E-Low			
Elective Paper	INTRODUCTION TO LINEAR ALGEBRA – PRACTICAL (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAGEP 05	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Acquire knowledge about Matrices and Caley-Hamilton Theorem.		
2	Understand the concepts of Symmetric and Skew Symmetric.		
	Course content		
Unit-I	Definitions of matrices – Addition, Subtraction and Multiplication of Matrices – Problems only.		
Unit-II	Transpose of a Matrix – Adjoint of a Matrix – Inverse of a matrix – Problems only.		
Unit-III	Definitions of Symmetric, Skew Symmetric, Hermitian and Skew Hermitian matrices – Problems only.		
Unit-IV	Rank of a Matrix: Definition – Finding the rank of Matrix – Problem upto 3 X 3 Matrix only.		
Unit-V	Characteristic equation of Matrix – Cayley Hamilton Theorem – Verification of Caley Hamilton Theorem – Simple Problems only.		
Text Books	1. Dr.P.R. Vittal, Allied Mathematics, Margham Publication, Chennai – 17, Reprint 2012.		
Reference Books	1. S.G.Venkatachalapathi, Allied Mathematics, Margham Publication, Chennai – 17, Reprint 2011.		
Web Resources	https://nptel.ac.in		
Pedagogy			
Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.			
Course Designer (Name & Signature)		Verified by BoS Chairman	

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	H	H	H	H	H	L	H	-	-
CO2	H	H	M	M	M	M	M	H	M	-
CO3	H	H	L	L	L	H	L	H	L	-
CO4	H	M	L	H	H	H	M	H	-	-
CO5	H	M	M	M	H	H	L	H	-	-

H-High; M-Medium; L-Low

Elective Paper	GRAPH THEORY AND ITS APPLICATION -PRACTICAL (FOR ALL COMPUTER SCIENCE DEPARTMENTS) ELECTIVE COURSE GENERIC SPECIFIC		
Course code	Total Hours	Contact Hours	Credits
25UMAGE P06	2	2	2
	Course Objectives		
	The main objectives of this course are		
1	Acquire knowledge about Graphs and Subgraphs.		
2	Understand the concepts of Walks, Trails and Paths.		
	Course content		
Unit-I	Graphs – Subgraphs – Operations on Graphs.		
Unit-II	Connected Graphs – Disconnected Graphs and Components.		
Unit-III	Walks, Trails and Paths.		
Unit-IV	Eulerian Graphs – Hamiltonian Graphs.		
Unit-V	Trees – Characterization of Trees – Centre of a Tree.		
Text Books	1. S.Arumugam, S. Ramachandran, Invitation to graph theory, Scitech Publications, Chennai, 2001.		
Reference Books	1. John clark and Derek Allan Hoton, A first book at graph theory, Allied publishes. 2. Harry, Graph Theory, Narosa Publications, New Delhi.		
Web Resources	https://nptel.ac.in		
Pedagogy	Lecture by Chalk and talk, Power point presentation, E-content, Simulation, Group discussion, Assignment, quiz and Seminar.		
Course Designer (Name & Signature)		Verified by BoS Chairman	

MAPPING WITH PROGRAMME OUTCOMES:

CO	PSOs					POs				
	1	2	3	4	5	1	2	3	4	5
CO1	H	L	H	H	H	H	L	H	-	-
CO2	M	M	M	H	M	M	M	H	M	-
CO3	L	H	L	L	L	H	L	H	L	-
CO4	M	H	L	M	H	H	M	H	-	-
CO5	M	M	M	M	H	H	L	H	-	-

H-High; **M**-Medium; **L**-Low